

**INTELLIGENT SYSTEMS AND DECISION SUPPORT SYSTEMS
PRACTICUM EVEN SEMESTER A/Y 2025/2026**

LAPORAN PROYEK AKHIR



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APPROVAL PAGE
FINAL PROJECT REPORT

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FOREWORD

I give thanks to God Almighty, who has always bestowed His grace and guidance, enabling me to complete the Intelligent Systems and Decision Support Systems practicum as well as the final practicum project report titled “A Decision Support System for Real Estate Selection in California Based on User Preferences Using the AHP Method.” This report details the final project I selected based on what I learned during the practicum.

I would also like to express my gratitude to the lab assistants who consistently guided and taught me throughout the lab and in the preparation of this report. This report is far from perfect; therefore, I welcome constructive criticism and suggestions to improve this final report.

I extend my thanks to everyone who assisted in the preparation of this report. I hope this report will be of use as needed.

Yogyakarta, 25 Mei 2026

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CHAPTER I

INTRODUCTION

1.1 Latar Belakang Masalah

Choosing a house is one of the important decisions that requires many considerations. Each prospective buyer has different preferences, such as house price, building area, number of bedrooms, number of bathrooms, and house age. The large number of available housing alternatives often makes the decision-making process difficult and time-consuming.

In addition, manual house selection processes often lead to less optimal decisions because buyers must compare a large amount of data simultaneously. Therefore, a Decision Support System (DSS) is needed to help users determine the best house based on their desired criteria.

In this final project, the Analytical Hierarchy Process (AHP) method is used because it can determine the priority level of each criterion through pairwise comparison. Using the AHP method, the system can provide more objective recommendations according to user preferences.

This system is implemented using the Python programming language with the Streamlit framework to create an interactive and user-friendly application.

1.2 Objectives

The objectives of this final project are:

1. To develop a web-based Decision Support System for house selection using Streamlit.
2. To implement the Analytical Hierarchy Process (AHP) method in the decision-making process.
3. To assist users in determining the best house based on multiple criteria.
4. To automatically display house ranking results based on user priority weights.

1.3 Benefits

The benefits of this final project are:

1. Helping users choose houses more quickly and effectively.
2. Reducing decision-making errors in house selection.
3. Providing house recommendations based on user priority needs.
4. Serving as a practical implementation of the AHP method in Decision Support Systems.

CHAPTER II DISCUSSION

2.1 Theoretical Foundation

a. Decision Support System (DSS)

A Decision Support System is a computer-based system used to assist the decision-making process for specific problems. DSS aims to provide recommendations based on data and certain calculation methods.

b. Analytical Hierarchy Process (AHP)

Analytical Hierarchy Process (AHP) is a decision-making method developed by Thomas L. Saaty. This method is used to determine priorities among alternatives based on several criteria.

The steps of the AHP method include:

- 1) Determining the criteria.
- 2) Creating a pairwise comparison matrix.
- 3) Calculating matrix normalization.
- 4) Calculating priority weights.
- 5) Calculating the Consistency Ratio (CR).
- 6) Determining the ranking of alternatives.

AHP uses a comparison scale from 1–9 to determine the importance level between criteria.

Nilai	Keterangan
1	Equally Important
3	Slightly More Important
5	More Important
7	Very Important
9	Absolutely More Important

2.2 Case Scenario & Dataset

In this project, users want to find the best house based on several important criteria. The system helps users determine the best house using the AHP method.

The criteria used are:

1. House Price
2. Building Area
3. Number of Bedrooms

4. Number of Bathrooms
5. House Age
6. Garage Space
7. Home Type

Dataset

The dataset used is the Real Estate California dataset containing residential property data in California, United States.

Some of the attributes used include:

Atribut	Keterangan
price	House price
livingArea	Building area
bedrooms	Number of bedrooms
bathrooms	Number of bathrooms
yearBuilt	Construction year
garageSpaces	Garage Spaces
homeType	Type of house
latitude	Latitude coordinate
longitude	Longitude coordinate

2.3 Decision Support System Modeling

Criteria and Types

Criteria	Type
House Price	Cost
Building Area	Benefit
Number of Bedrooms	Benefit
Number of Bathrooms	Benefit
House Age	Cost
Home Type	Benefit
Garage Space	Benefit

Alternatives

The alternatives used are houses from the Real Estate California dataset.

Pairwise Comparison

Users compare criteria using the AHP scale from 1–9.

Perbandingan	Nilai
Price vs Building Area	5
Price vs Bedrooms	3
Price vs Bathrooms	3

2.4 Calculation and Algorithm

AHP Calculation Steps

1. The pairwise matrix is used to compare the importance level between criteria.
2. Normalisasi Matriks

Each value in the matrix is divided by the total of its respective column.

$$normalization = \frac{value}{sum\ of\ column}$$

3. Calculating Priority Weights

Priority weights are obtained from the average of each row in the normalized matrix.

$$Bobot = \frac{jumlah\ baris}{jumlah\ kriteria}$$

4. Consistency Calculation

Consistency is calculated using the Consistency Ratio(CR):

$$CI = \frac{\lambda_{max} - n}{n-1} CR = \frac{CI}{RI}$$

If:

$$CR < 0.1$$

then the matrix is considered consistent.

5. Alternative Ranking

After obtaining the weights, the system calculates the score of each house based on AHP weights and criterion values. The alternative with the highest score becomes the main recommendation.

2.5 Implementasi & Tampilan Sistem

```
import streamlit as st
import pandas as pd
import numpy as np
import base64
import plotly.express as px

# =====
# 1. PENGATURAN KONFIGURASI HALAMAN UTAMA & BACKEND MEMORI
# =====
st.set_page_config(page_title="SPK Pemilihan Rumah - 7 Kriteria AHP",
layout="centered")

if 'hasil_terakhir' not in st.session_state:
    st.session_state.hasil_terakhir = None

# =====
# 2. IMPLEMENTASI BACKGROUND & PRESET CSS CUSTOMIZATION CONTRAST
# =====
def get_base64_of_bin_file(bin_file):
    with open(bin_file, 'rb') as f:
        data = f.read()
    return base64.b64encode(data).decode()

def set_bg_from_local(gambar_file):
    try:
        bin_str = get_base64_of_bin_file(gambar_file)
```

```

        bg_css = f'background-image:
url("data:image/png;base64,{bin_str}");'
    except Exception:
        bg_css = 'background: linear-gradient(135deg, #0f2027, #203a43,
#2c5364);'

    page_bg_img = f'''
<style>

                                                                    @import
url('https://fonts.googleapis.com/css2?family=Plus+Jakarta+Sans:wght=500;
700;800&display=swap');

.stApp {{
    {bg_css}
    background-attachment: fixed;
    background-size: cover;
    font-family: 'Plus Jakarta Sans', sans-serif !important;
    text-align: center;
}}

.stMarkdown p, .stMarkdown li {{
    text-align: center !important;
}}

.stMarkdown, .stSlider, .stSelectbox, .stTextInput, .stTable,
div[data-testid="stAlert"], .stRadio, div.row-widget.stRadio > div {{
    background-color: rgba(15, 32, 39, 0.85);
    padding: 20px;
    border-radius: 12px;
    margin-bottom: 15px;
    color: #ffffff !important;
    font-weight: 700 !important;
    box-shadow: 0px 8px 24px rgba(0, 0, 0, 0.4);
    border: 2px solid #00f0ff !important;
}}

div[data-testid="stAlert"] {{
    border: 2px solid #00f0ff !important;
    background-color: rgba(15, 32, 39, 0.9) !important;
}}

.stApp label, .stApp p, .stApp small {{
    color: #ffffff !important;
    font-weight: 700 !important;
    justify-content: center;
}}

/* CSS DROPDOWN KONTRAS FIX */

                                                                    .stApp
div:not([data-baseweb="select"]):not([data-baseweb="popover"]):not([role=
"option"]):not([data-baseweb="menu"]),

                                                                    .stApp
span:not([data-baseweb="select"]):not([data-baseweb="popover"]):not([role
="option"]):not([data-baseweb="menu"]) {{
    color: #ffffff;
}}

h1, h2, h3, h4 {{
    background-color: rgba(15, 32, 39, 0.9);

```

```

padding: 15px;
border-radius: 12px;
color: #ffffff !important;
font-weight: 800 !important;
text-align: center !important;
border: 2px solid #00f0ff !important;
border-bottom: 6px solid #00f0ff !important;
box-shadow: 0px 8px 20px rgba(0, 0, 0, 0.3);
}}

hr {{
border: 0;
height: 3px;
background: linear-gradient(to right, transparent, #00f0ff,
transparent);
margin: 25px 0;
}}

div[data-testid="stTextInput"] input, div[data-testid="stSelectbox"]
div[data-baseweb="select"] {{
background-color: #ffffff !important;
border: 2px solid #00f0ff !important;
border-radius: 8px !important;
}}

div[data-baseweb="select"] div,
div[data-baseweb="select"] span,
div[data-baseweb="select"] [aria-selected="true"] {{
color: #000000 !important;
font-weight: 700 !important;
-webkit-text-fill-color: #000000 !important;
}}

div[data-testid="stTextInput"] input {{
color: #000000 !important;
font-weight: 700 !important;
-webkit-text-fill-color: #000000 !important;
}}

div[data-baseweb="popover"] ul,
div[data-baseweb="popover"] li,
div[data-baseweb="popover"] div,
div[data-baseweb="menu"] div,
div[role="option"],
div[role="option"] span,
div[role="option"] div {{
background-color: #ffffff !important;
color: #000000 !important;
font-weight: 700 !important;
-webkit-text-fill-color: #000000 !important;
}}

div[data-testid="stSlider"] div[role="slider"] {{
background-color: #00f0ff !important;
border: 2px solid #ffffff !important;
box-shadow: 0px 0px 8px #00f0ff;
}}
div[data-testid="stSlider"] div[aria-hidden="true"] > div {{
background-color: #00f0ff !important;

```

```

    }}
    div[data-testid="stSlider"] div[aria-hidden="true"] > div:first-child
  {{
    background-color: rgba(255, 255, 255, 0.2) !important;
  }}

  div[data-testid="stTable"] table, div[data-testid="stDataFrame"] div
  {{
    border: 2px solid #00f0ff !important;
    color: #ffffff !important;
  }}

  div.stButton > button:first-child {{
    background-color: #00f0ff !important;
    color: #000000 !important;
    font-weight: 800 !important;
    border: 2px solid #000000 !important;
    border-radius: 8px !important;
    padding: 0.6rem 2.5rem !important;
    transition: all 0.3s cubic-bezier(0.25, 0.8, 0.25, 1);
    box-shadow: 0px 4px 12px rgba(0, 240, 255, 0.4);
    width: 100%;
  }}

  div.stButton > button:first-child:hover {{
    background-color: #00e5ff !important;
    box-shadow: 0px 8px 20px rgba(0, 240, 255, 0.7);
    transform: translateY(-3px);
  }}
</style>
'''

st.markdown(page_bg_img, unsafe_allow_html=True)

set_bg_from_local('pexels-stephen-leonardi-587681991-34276128.jpg')

# =====
# 3. ENGINE MEMUAT DATASET UTAMA (7 KRITERIA DASAR)
# =====
@st.cache_data
def load_data():
    df = pd.read_csv('RealEstate_California.csv')
    df_clean = df[df['homeType'].isin(['SINGLE_FAMILY', 'CONDO',
'TOWNHOUSE'])].copy()

    if 'garageSpaces' in df_clean.columns:
        df_clean['garageSpaces'] = df_clean['garageSpaces'].fillna(0)
    else:
        df_clean['garageSpaces'] = 0

    df_clean = df_clean.dropna(subset=['price', 'livingArea', 'bedrooms',
'bathrooms', 'yearBuilt'])

    df_clean = df_clean[(df_clean['price'] > 0) &
                        (df_clean['livingArea'] > 0) &
                        (df_clean['bedrooms'] > 0) &
                        (df_clean['bathrooms'] > 0) &
                        (df_clean['yearBuilt'] > 0)]

    df_clean['house_age'] = 2021 - df_clean['yearBuilt']

```

```

df_clean['house_age'] = df_clean['house_age'].replace(0, 1)

return df_clean

try:
    df_rumah = load_data()
except Exception as e:
    st.error(f"Gagal memuat file RealEstate_California.csv. Error: {e}")
    st.stop()

# Skema Bobot Standar Intensitas Penilaian Absolut (Skor Lokal Alternatif)
bobot_rating = {
    'price': {'Murah': 0.65, 'Sedang': 0.25, 'Mahal': 0.10},
    'livingArea': {'Luas': 0.60, 'Cukup': 0.30, 'Sempit': 0.10},
    'bedrooms': {'Banyak': 0.63, 'Standar': 0.26, 'Sedikit': 0.11},
    'bathrooms': {'Banyak': 0.63, 'Standar': 0.26, 'Sedikit': 0.11},
    'house_age': {'Baru': 0.65, 'Sedang': 0.25, 'Tua': 0.10},
    'homeType': {'SINGLE_FAMILY': 0.60, 'TOWNHOUSE': 0.30, 'CONDO': 0.10},
    'garageSpaces': {'Ada Garasi Luas': 0.65, 'Ada Garasi Standar': 0.25, 'Tidak Ada Garasi': 0.10}
}

kriteria_label = [
    'Harga Rumah (C1)', 'Luas Bangunan (C2)', 'Jumlah Kamar Tidur (C3)',
    'Jumlah Kamar Mandi (C4)', 'Umur Fisik Rumah (C5)', 'Jenis Rumah (C6)', 'Fasilitas Garasi (C7)'
]

# =====
# 4. SISTEM MENU NAVIGASI (SIDEBAR)
# =====
st.sidebar.title("Navigasi Aplikasi")
menu_halaman = st.sidebar.radio(
    "Pilih Halaman:",
    ["Halaman Profil Kelompok", "Halaman Data", "Halaman Hitung SPK", "Halaman Visualisasi Peta"]
)

# -----
# HALAMAN 1: PROFIL KELOMPOK
# -----
if menu_halaman == "Halaman Profil Kelompok":
    st.title("Halaman Profil Kelompok")
    st.write("Sistem Pendukung Keputusan Pemilihan Properti Rumah Ideal Menggunakan Metode AHP.")

    st.info("""
    Anggota Kelompok / Pengembang:
    1. Kevin Ridoi Parhusip (NIM. 123240084) - Informatika H
    2. Oktavian Prasetya Adi (NIM. 123240135) - Informatika H
    """)
    st.write("Silakan gunakan menu navigasi di Sidebar sebelah kiri untuk berpindah halaman.")

# -----
# HALAMAN 2: DATA EXPLORATION
# -----

```



```

elif menu_halaman == "Halaman Data":
    st.title("Eksplorasi Dataset Mentah")
    st.write("Menampilkan data komprehensif real estate wilayah California hasil unduhan online.")

    st.success(f"Syarat Jumlah Baris Terpenuhi: Dataset memiliki {df_rumah.shape[0]} baris data (Minimal syarat: 250 baris).")
    st.success("Syarat Jumlah Kriteria Terpenuhi: Memiliki 7 Kriteria Utama (Harga, Luas, Kamar Tidur, Kamar Mandi, Umur Rumah, Jenis Rumah, Garasi).")

    st.subheader("Tabel Interaktif Dataset Mentah (st.dataframe)")
    st.dataframe(df_rumah[['streetAddress', 'city', 'price', 'livingArea', 'bedrooms', 'bathrooms', 'house_age', 'homeType', 'garageSpaces']], use_container_width=True)

# -----
# HALAMAN 3: PROSES KOMPUTASI & HITUNG SPK (10 PERSONAL BUYER)
# -----
elif menu_halaman == "Halaman Hitung SPK":
    st.title("Perhitungan SPK (Metode AHP - 7 Kriteria)")
    st.write("Sesuaikan tingkat prioritas komparasi berpasangan antar kriteria.")

    st.subheader("Input Bobot & Filter Dinamis")
    kota_pilihan = st.selectbox("Saring Kota Penempatan Properti:", ['Semua Kota'] + sorted(df_rumah['city'].unique().tolist()))

    # INDIKATOR INPUT PAIRWISE DENGAN 10 TEMPLATE BUYER PERSONA
    st.write("Prosedur Pengisian Matriks Perbandingan Berpasangan:")
    metode_input = st.radio(
        "Pilih Metode Pengisian Preferensi:",
        ["Gunakan Template Buyer Properti (10 Persona)", "Isi Manual via Slider"]
    )

    matriks_ahp = np.ones((7, 7))

    if metode_input == "Gunakan Template Buyer Properti (10 Persona)":
        tipe_buyer = st.selectbox(
            "Pilih Profil Calon Pembeli (Buyer Persona):",
            [
                "1. Keluarga Besar (Prioritas Kamar Banyak & Luas Bangunan)",
                "2. Pasangan Suami Istri Baru (Prioritas Finansial & Harga Ekonomis)",
                "3. Pekerja Lajang / Eksekutif (Prioritas Garasi & Rumah Minimalis)",
                "4. Investor Real Estate (Prioritas Harga Murah & Umur Rumah Baru)",
                "5. Keluarga Muda Beranak Satu (Prioritas Keseimbangan Harga & Kamar)",
                "6. Lansia / Pensiunan (Prioritas Rumah Tua/Klasik & Luas Halaman)",
                "7. Kolektor Mobil / Otomotif (Prioritas Utama Garasi Sangat Luas)",
                "8. Pengusaha Home Industry (Prioritas Luas Bangunan Maksimal)",
            ]
        )

```

```

"9. Pemburu Rumah Baru / Gress (Prioritas Umur Fisik
Rumah Terkecil)",
"10. Kaum Urban / Praktis (Prioritas
Kondominium/Townhouse Modern)"
]
)

# LOGIKA MATRIKS BERDASARKAN 10 BUYER PERSONA (PASTI KONSISTEN <
0.1)
if "1. Keluarga Besar" in tipe_buyer:
    st.info("Template Aktif: Mengutamakan kapasitas ruangan
(Kamar Tidur/Mandi) dan kelonggaran Luas Bangunan.")
    c12, c13, c14, c15, c16, c17 = 1, 0.33, 0.33, 2, 1, 2
    c23, c24, c25, c26, c27 = 1, 1, 4, 3, 3
    c34, c35, c36, c37 = 2, 5, 4, 4
    c45, c46, c47 = 4, 3, 3
    c56, c57 = 0.5, 0.5
    c67 = 1

elif "2. Pasangan Suami Istri Baru" in tipe_buyer:
    st.info("Template Aktif: Fokus menekan budget pengeluaran
awal (Harga Rumah C1 sangat dominan).")
    c12, c13, c14, c15, c16, c17 = 4, 6, 6, 5, 4, 5
    c23, c24, c25, c26, c27 = 2, 2, 3, 1, 2
    c34, c35, c36, c37 = 1, 1, 0.5, 1
    c45, c46, c47 = 1, 0.5, 1
    c56, c57 = 0.5, 1
    c67 = 2

elif "3. Pekerja Lajang" in tipe_buyer:
    st.info("Template Aktif: Mengutamakan ketersediaan Fasilitas
Garasi (C7) dan fleksibilitas kendaraan.")
    c12, c13, c14, c15, c16, c17 = 1, 3, 3, 1, 0.33, 0.25
    c23, c24, c25, c26, c27 = 2, 2, 1, 0.5, 0.33
    c34, c35, c36, c37 = 1, 0.5, 0.33, 0.2
    c45, c46, c47 = 1, 0.33, 0.2
    c56, c57 = 0.5, 0.33
    c67 = 0.5

elif "4. Investor Real Estate" in tipe_buyer:
    st.info("Template Aktif: Fokus memaksimalkan margin
keuntungan lewat Harga Beli Rendah (C1) dan bangunan awet.")
    c12, c13, c14, c15, c16, c17 = 5, 5, 5, 1, 4, 4
    c23, c24, c25, c26, c27 = 1, 1, 0.25, 2, 1
    c34, c35, c36, c37 = 1, 0.2, 1, 1
    c45, c46, c47 = 0.2, 1, 1
    c56, c57 = 4, 3
    c67 = 1

elif "5. Keluarga Muda Beranak Satu" in tipe_buyer:
    st.info("Template Aktif: Keseimbangan proporsional antara
Harga Terjangkau dan Kamar Tidur Standar.")
    c12, c13, c14, c15, c16, c17 = 2, 2, 3, 2, 2, 3
    c23, c24, c25, c26, c27 = 1, 2, 1, 1, 2
    c34, c35, c36, c37 = 2, 1, 1, 2
    c45, c46, c47 = 0.5, 0.5, 1
    c56, c57 = 1, 2
    c67 = 1

```

```

        elif "6. Lansia / Pensiunan" in tipe_buyer:
            st.info("Template Aktif: Mengabaikan tren modernitas, menyukai rumah mapan/umur matang yang tenang.")
            c12, c13, c14, c15, c16, c17 = 1, 2, 2, 0.25, 2, 1
            c23, c24, c25, c26, c27 = 2, 2, 0.2, 1, 1
            c34, c35, c36, c37 = 1, 0.2, 0.5, 0.5
            c45, c46, c47 = 0.2, 0.5, 0.5
            c56, c57 = 4, 3
            c67 = 1

        elif "7. Kolektor Mobil / Otomotif" in tipe_buyer:
            st.info("Template Aktif: Menempatkan Fasilitas Garasi (C7) sebagai kriteria mutlak di atas segalanya.")
            c12, c13, c14, c15, c16, c17 = 1, 1, 1, 1, 1, 1
            c23, c24, c25, c26, c27 = 1, 1, 1, 1, 1
            c34, c35, c36, c37 = 1, 1, 1, 1
            c45, c46, c47 = 1, 1, 1
            c56, c57 = 1, 1
            c67 = 1
            # Override manual untuk menjamin dominasi mutlak C7 secara konsisten
            matriks_ahp[:, 6] = [8, 7, 5, 5, 7, 7, 1]
            matriks_ahp[6, :] = [1/8, 1/7, 1/5, 1/5, 1/7, 1/7, 1]

        elif "8. Pengusaha Home Industry" in tipe_buyer:
            st.info("Template Aktif: Luas Bangunan (C2) sangat dominan untuk ruang produksi/gudang penyimpanan.")
            c12, c13, c14, c15, c16, c17 = 0.25, 0.2, 0.2, 1, 1, 1
            c23, c24, c25, c26, c27 = 4, 4, 5, 4, 5
            c34, c35, c36, c37 = 1, 2, 1, 2
            c45, c46, c47 = 2, 1, 2
            c56, c57 = 0.5, 1
            c67 = 2

        elif "9. Pemburu Rumah Baru" in tipe_buyer:
            st.info("Template Aktif: Memprioritaskan faktor Fisik Umur Rumah (C5) sekecil mungkin agar bebas renovasi.")
            c12, c13, c14, c15, c16, c17 = 1, 2, 2, 0.2, 1, 2
            c23, c24, c25, c26, c27 = 2, 2, 0.14, 1, 1
            c34, c35, c36, c37 = 1, 0.14, 0.5, 1
            c45, c46, c47 = 0.14, 0.5, 1
            c56, c57 = 5, 6
            c67 = 1

        else: # 10. Kaum Urban / Praktis
            st.info("Template Aktif: Mengutamakan Jenis Properti (C6) spesifik seperti Townhouse/Condo tengah kota demi efisiensi.")
            c12, c13, c14, c15, c16, c17 = 1, 2, 2, 1, 0.2, 1
            c23, c24, c25, c26, c27 = 1, 1, 1, 0.14, 1
            c34, c35, c36, c37 = 1, 1, 0.14, 1
            c45, c46, c47 = 1, 0.14, 1
            c56, c57 = 5, 6
            c67 = 0.2

    else:
        st.info("Atur parameter 21 komponen slider di bawah ini secara manual:")
        c12 = st.slider("Harga (C1) VS Luas Bangunan (C2)", 1, 9, 3)
        c13 = st.slider("Harga (C1) VS Jumlah Kamar Tidur (C3)", 1, 9, 5)

```

```

c14 = st.slider("Harga (C1) VS Jumlah Kamar Mandi (C4)", 1, 9, 5)
c15 = st.slider("Harga (C1) VS Umur Rumah (C5)", 1, 9, 4)
c16 = st.slider("Harga (C1) VS Jenis Rumah (C6)", 1, 9, 3)
c17 = st.slider("Harga (C1) VS Fasilitas Garasi (C7)", 1, 9, 4)

c23 = st.slider("Luas Bangunan (C2) VS Jumlah Kamar Tidur (C3)",
1, 9, 3)
c24 = st.slider("Luas Bangunan (C2) VS Jumlah Kamar Mandi (C4)",
1, 9, 3)
c25 = st.slider("Luas Bangunan (C2) VS Umur Rumah (C5)", 1, 9, 2)
c26 = st.slider("Luas Bangunan (C2) VS Jenis Rumah (C6)", 1, 9,
2)
c27 = st.slider("Luas Bangunan (C2) VS Fasilitas Garasi (C7)", 1,
9, 3)

c34 = st.slider("Jumlah Kamar Tidur (C3) VS Jumlah Kamar Mandi
(C4)", 1, 9, 1)
c35 = st.slider("Jumlah Kamar Tidur (C3) VS Umur Rumah (C5)", 1,
9, 1)
c36 = st.slider("Jumlah Kamar Tidur (C3) VS Jenis Rumah (C6)", 1,
9, 2)
c37 = st.slider("Jumlah Kamar Tidur (C3) VS Fasilitas Garasi
(C7)", 1, 9, 2)

c45 = st.slider("Jumlah Kamar Mandi (C4) VS Umur Rumah (C5)", 1,
9, 1)
c46 = st.slider("Jumlah Kamar Mandi (C4) VS Jenis Rumah (C6)", 1,
9, 1)
c47 = st.slider("Jumlah Kamar Mandi (C4) VS Fasilitas Garasi
(C7)", 1, 9, 2)

c56 = st.slider("Umur Rumah (C5) VS Jenis Rumah (C6)", 1, 9, 1)
c57 = st.slider("Umur Rumah (C5) VS Fasilitas Garasi (C7)", 1, 9,
1)

c67 = st.slider("Jenis Rumah (C6) VS Fasilitas Garasi (C7)", 1,
9, 1)

# Sinkronisasi pemetaan data ke dalam matriks berpasangan beraturan
resiprokal
if metode_input != "Gunakan Template Buyer Properti (10 Persona)" or
"7. Kolektor Mobil" not in tipe_buyer:
    matriks_ahp[0, 1] = c12; matriks_ahp[1, 0] = 1 / c12
    matriks_ahp[0, 2] = c13; matriks_ahp[2, 0] = 1 / c13
    matriks_ahp[0, 3] = c14; matriks_ahp[3, 0] = 1 / c14
    matriks_ahp[0, 4] = c15; matriks_ahp[4, 0] = 1 / c15
    matriks_ahp[0, 5] = c16; matriks_ahp[5, 0] = 1 / c16
    matriks_ahp[0, 6] = c17; matriks_ahp[6, 0] = 1 / c17

    matriks_ahp[1, 2] = c23; matriks_ahp[2, 1] = 1 / c23
    matriks_ahp[1, 3] = c24; matriks_ahp[3, 1] = 1 / c24
    matriks_ahp[1, 4] = c25; matriks_ahp[4, 1] = 1 / c25
    matriks_ahp[1, 5] = c26; matriks_ahp[5, 1] = 1 / c26
    matriks_ahp[1, 6] = c27; matriks_ahp[6, 1] = 1 / c27

    matriks_ahp[2, 3] = c34; matriks_ahp[3, 2] = 1 / c34
    matriks_ahp[2, 4] = c35; matriks_ahp[4, 2] = 1 / c35
    matriks_ahp[2, 5] = c36; matriks_ahp[5, 2] = 1 / c36
    matriks_ahp[2, 6] = c37; matriks_ahp[6, 2] = 1 / c37

```

```

matriks_ahp[3, 4] = c45; matriks_ahp[4, 3] = 1 / c45
matriks_ahp[3, 5] = c46; matriks_ahp[5, 3] = 1 / c46
matriks_ahp[3, 6] = c47; matriks_ahp[6, 3] = 1 / c47

matriks_ahp[4, 5] = c56; matriks_ahp[5, 4] = 1 / c56
matriks_ahp[4, 6] = c57; matriks_ahp[6, 4] = 1 / c57

matriks_ahp[5, 6] = c67; matriks_ahp[6, 5] = 1 / c67

st.write("---")
jalankan_hitung = st.button("Eksekusi Perhitungan Matriks &
Perangkingan")

if jalankan_hitung:
    # Komputasi Nilai Bobot Eigenvector Kriteria Utama
    kolom_sum = matriks_ahp.sum(axis=0)
    matriks_normalisasi = matriks_ahp / kolom_sum
    bobot_kriteria = matriks_normalisasi.mean(axis=1)

    # Validasi Uji Konsistensi Logika Matematis (CR Saaty Ordo 7)
    RI = 1.32
    weighted_sum = np.dot(matriks_ahp, bobot_kriteria)
    consistency_vector = weighted_sum / bobot_kriteria
    lambda_max = np.mean(consistency_vector)
    CI = (lambda_max - 7) / 6
    CR = CI / RI if CI > 0 else 0

    if CR < 0.1:
        st.success(f"Preferensi Pengisian Konsisten! (Nilai CR =
{CR:.4f})")
    else:
        st.warning(f"Pola Kurang Konsisten, disarankan meninjau ulang
susunan slider (Nilai CR = {CR:.4f})")

    # PROSES TAMPILAN 3 TAHAP VISUALISASI MATRIKS UTAMA
    st.subheader("Visualisasi Tahapan Perhitungan Matriks AHP")

    st.write("1. Matriks Perbandingan Berpasangan Asli (Pairwise
Comparison Matrix):")
    df_view_pairwise = pd.DataFrame(matriks_ahp,
columns=kriteria_label, index=kriteria_label)
    st.dataframe(df_view_pairwise.style.format("{:.2f}"),
use_container_width=True)

    st.write("2. Matriks Perbandingan Berpasangan Ternormalisasi
Kolom:")
    df_view_normal = pd.DataFrame(matriks_normalisasi,
columns=kriteria_label, index=kriteria_label)
    st.dataframe(df_view_normal.style.format("{:.3f}"),
use_container_width=True)

    st.write("3. Hasil Akhir Vektor Bobot Kriteria Utama
(Eigenvector):")
    df_bobot = pd.DataFrame({
        'Kriteria Utama': kriteria_label,
        'Bobot Pengaruh': bobot_kriteria
    })

```

```

        st.dataframe(df_bobot.style.format({'Bobot Pengaruh': '{:.4f}'}),
use_container_width=True)

        # VISUALISASI 1: GRAFIK PEMBOBOTAN SEBARAN BOBOT KRITERIA UTAMA
        st.write("4. Grafik Distribusi Pembobotan Prioritas Kriteria:")
        df_grafik_bobot = df_bobot.sort_values(by='Bobot Pengaruh',
ascending=True)
        fig_bobot = px.bar(
            df_grafik_bobot, x='Bobot Pengaruh', y='Kriteria Utama',
orientation='h',
            title="Prioritas Kontribusi Bobot Pengaruh Kriteria",
            color='Bobot Pengaruh', color_continuous_scale='GnBu'
        )
        fig_bobot.update_layout(template="plotly_dark")
        st.plotly_chart(fig_bobot, use_container_width=True)

        # Penyaringan Data Berdasarkan Subset Kota Terpilih
        if kota_pilihan != 'Semua Kota':
            alternatif = df_rumah[df_rumah['city'] ==
kota_pilihan].copy()
        else:
            alternatif = df_rumah.head(1000).copy()

        if alternatif.empty:
            st.error("Properti tidak ditemukan pada kota yang dipilih.")
        else:
            # PROSES FILTER FILTER DINAMIS BERDASARKAN KUANTIL STATISTIK
DATA
            p_low, p_high = alternatif['price'].quantile([0.33, 0.66])
            l_low, l_high = alternatif['livingArea'].quantile([0.33,
0.66])
            a_low, a_high = alternatif['house_age'].quantile([0.33,
0.66])

            def get_r_price(x): return 'Murah' if x <= p_low else
('Sedang' if x <= p_high else 'Mahal')
            def get_r_area(x): return 'Sempit' if x <= l_low else
('Cukup' if x <= l_high else 'Luas')
            def get_r_beds(x): return 'Sedikit' if x <= 2 else ('Standar'
if x == 3 else 'Banyak')
            def get_r_baths(x): return 'Sedikit' if x <= 1.5 else
('Standar' if x <= 2.5 else 'Banyak')
            def get_r_age(x): return 'Baru' if x <= a_low else ('Sedang'
if x <= a_high else 'Tua')
            def get_r_garage(x): return 'Tidak Ada Garasi' if x == 0 else
('Ada Garasi Standar' if x <= 2 else 'Ada Garasi Luas')

            alternatif['r_price'] =
alternatif['price'].apply(get_r_price)
            alternatif['r_area'] =
alternatif['livingArea'].apply(get_r_area)
            alternatif['r_beds'] =
alternatif['bedrooms'].apply(get_r_beds)
            alternatif['r_baths'] =
alternatif['bathrooms'].apply(get_r_baths)
            alternatif['r_age'] =
alternatif['house_age'].apply(get_r_age)
            alternatif['r_garage'] =
alternatif['garageSpaces'].apply(get_r_garage)

```

```

def hitung_skor(row):
    s_price = bobot_kriteria[0] *
    bobot_rating['price'][row['r_price']]
    s_area = bobot_kriteria[1] *
    bobot_rating['livingArea'][row['r_area']]
    s_beds = bobot_kriteria[2] *
    bobot_rating['bedrooms'][row['r_beds']]
    s_baths = bobot_kriteria[3] *
    bobot_rating['bathrooms'][row['r_baths']]
    s_age = bobot_kriteria[4] *
    bobot_rating['house_age'][row['r_age']]
    s_type = bobot_kriteria[5] *
    bobot_rating['homeType'].get(row['homeType'], 0.10)
    s_garage = bobot_kriteria[6] *
    bobot_rating['garageSpaces'][row['r_garage']]
    return s_price + s_area + s_beds + s_baths + s_age +
    s_type + s_garage

alternatif['Skor Akhir'] = alternatif.apply(hitung_skor,
axis=1)

hasil_ranking = alternatif.sort_values(by='Skor Akhir',
ascending=False).head(10).reset_index()
hasil_ranking['ID Rumah'] = "Properti ID-" +
hasil_ranking['index'].astype(str)

tabel_final = hasil_ranking[['ID Rumah', 'streetAddress',
'city', 'price', 'livingArea', 'bedrooms', 'bathrooms', 'house_age',
'homeType', 'garageSpaces', 'Skor Akhir']].copy()
tabel_final.columns = ['ID Rumah', 'Alamat', 'Kota', 'Harga
(USD)', 'Luas (Sqft)', 'Kamar Tidur', 'Kamar Mandi', 'Umur Rumah', 'Jenis
Properti', 'Kapasitas Garasi', 'Skor Akhir']

st.subheader("Hasil Akhir Perangkingan (Top 10 Rekomendasi)")
st.dataframe(tabel_final.style.format({
    'Harga (USD)': '${:,.2f}',
    'Luas (Sqft)': '{:,.0f}',
    'Kapasitas Garasi': '{:.0f} mobil',
    'Skor Akhir': '{:.4f}'
}), use_container_width=True)

# VISUALISASI 2: GRAFIK KOMPARASI PEMERINGKATAN SKOR AKHIR
ALTERNATIF TOP 10
st.write("Grafik Pemeringkatan Nilai Skor Akhir Alternatif
Properti (Top 10):")
fig_skor = px.bar(
    tabel_final, x='ID Rumah', y='Skor Akhir',
    title="Peringkat Skor Akhir Akumulasi Global SPK AHP",
    color='Skor Akhir', color_continuous_scale='Viridis'
)
fig_skor.update_layout(template="plotly_dark")
st.plotly_chart(fig_skor, use_container_width=True)

st.session_state.hasil_terakhir = hasil_ranking
else:
    st.info("Atur tingkat kepentingan antar kriteria di atas,
kemudian tekan tombol 'Eksekusi Perhitungan Matriks & Perangkingan' untuk
melihat hasil.")

```

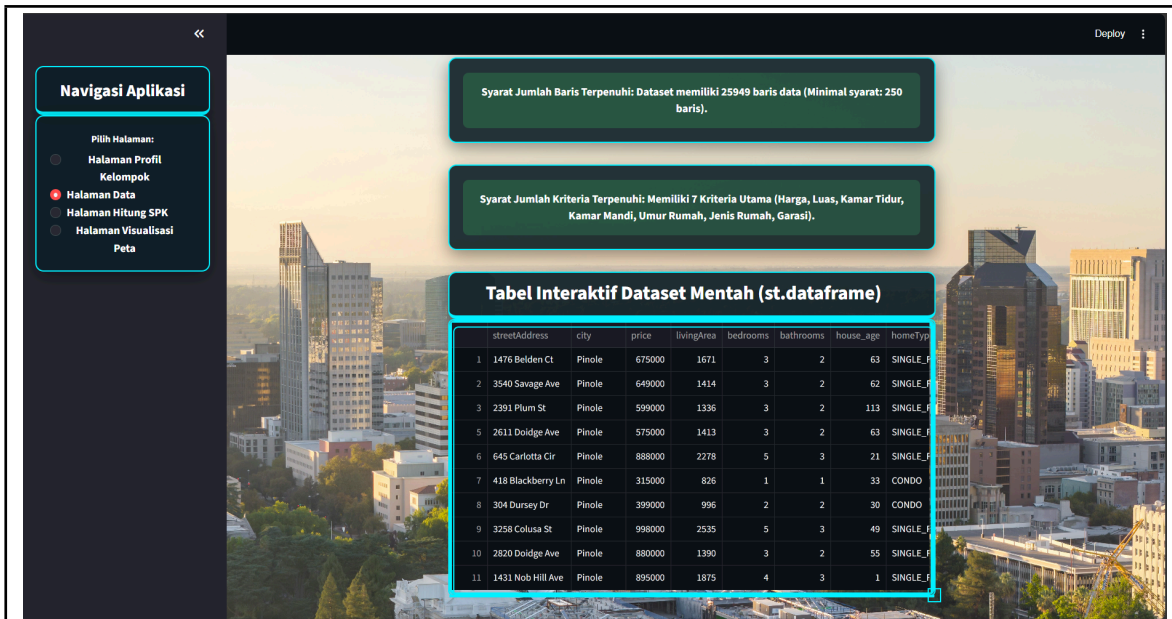
```
# -----
# HALAMAN 4: VISUALISASI SEBARAN SPASIAL (PETA)
# -----
elif menu_halaman == "Halaman Visualisasi Peta":
    st.title("Distribusi Geografis Properti")
    st.write("Halaman ini menyajikan peta sebaran alternatif rumah rekomendasi.")

    if st.session_state.hasil_terakhir is not None and not st.session_state.hasil_terakhir.empty:
        st.map(st.session_state.hasil_terakhir[['latitude', 'longitude']].dropna(), latitude='latitude', longitude='longitude')
    else:
        st.warning("Belum ada data perangkingan untuk ditampilkan. Silakan hitung terlebih dahulu di halaman 'Halaman Hitung SPK'.")
```

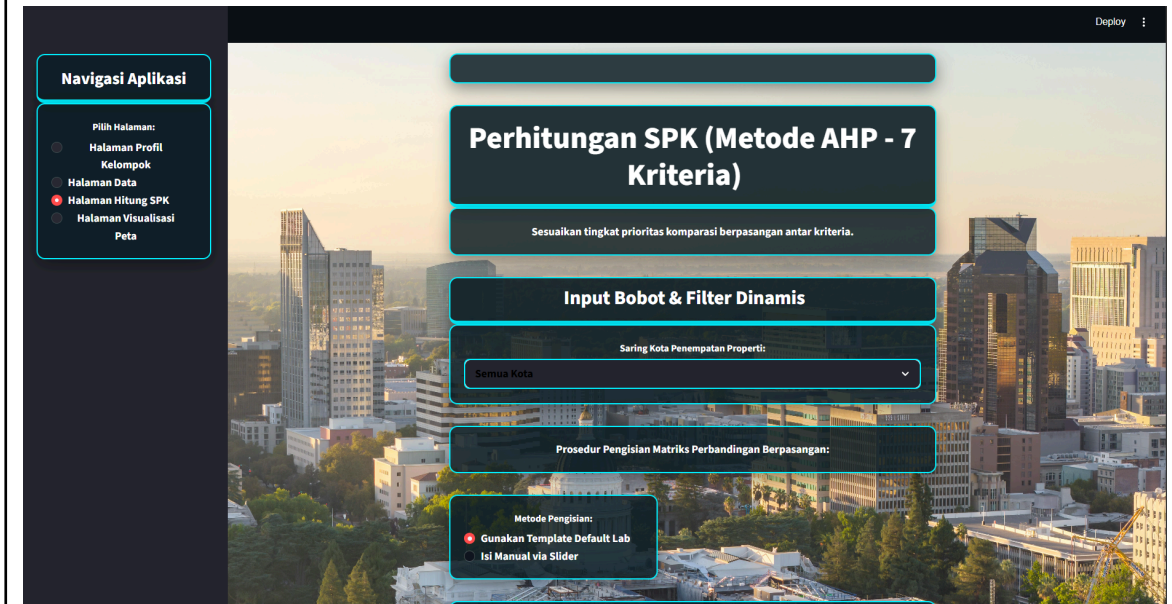
Picture 1 Landing and Profile Page



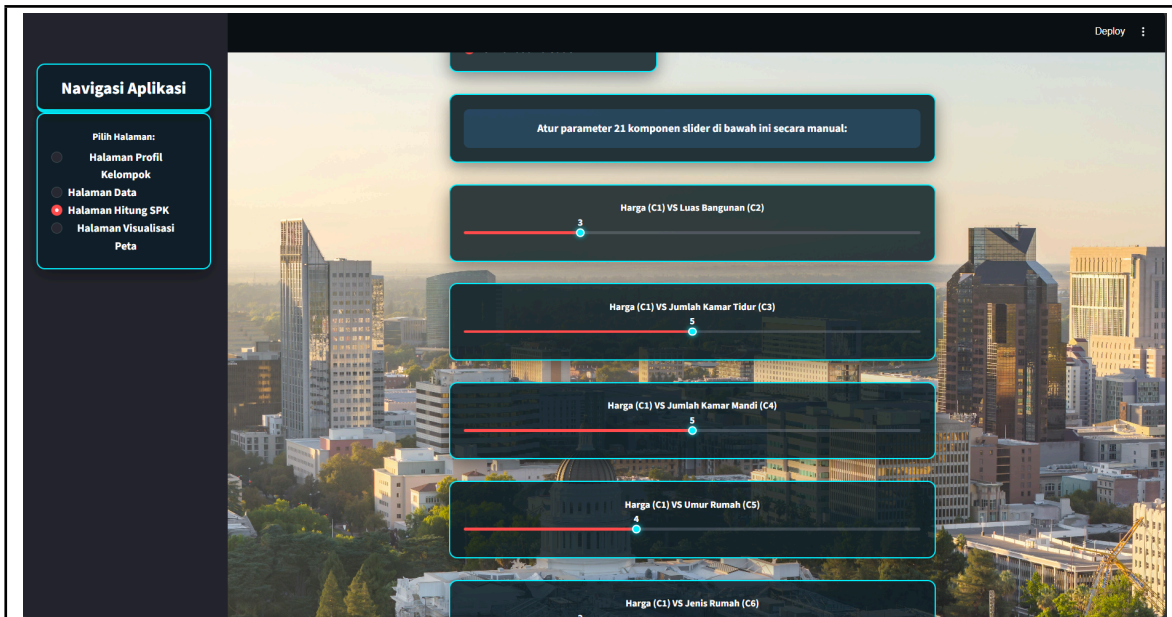
Picture 2 Dataset Page



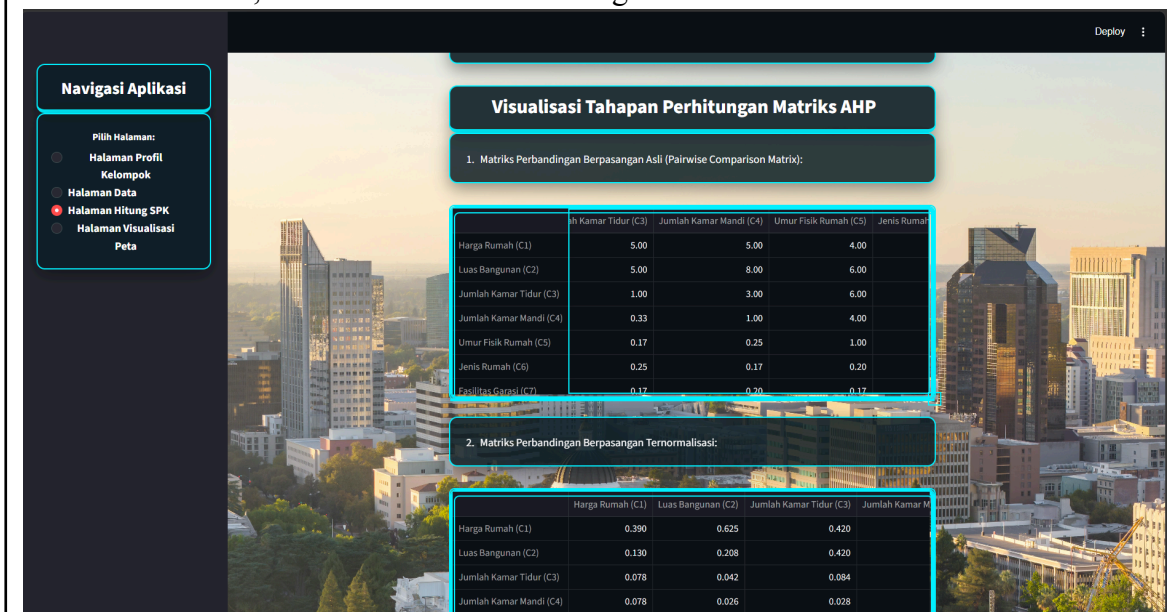
Picture 3 DSS



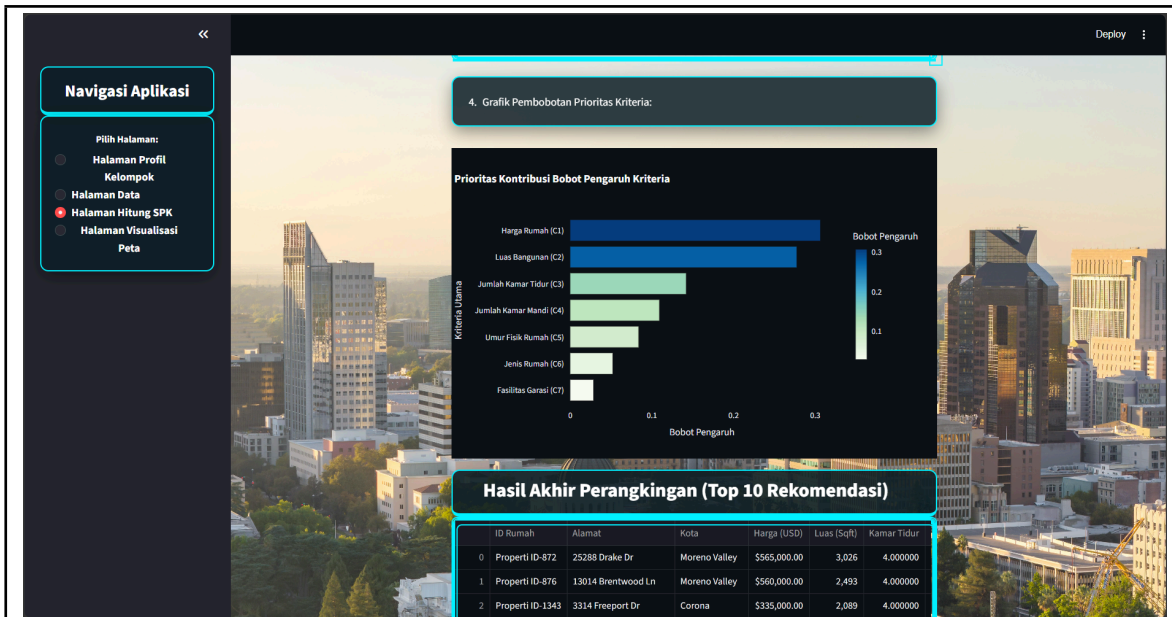
Picture 4 Filling in the level of importance



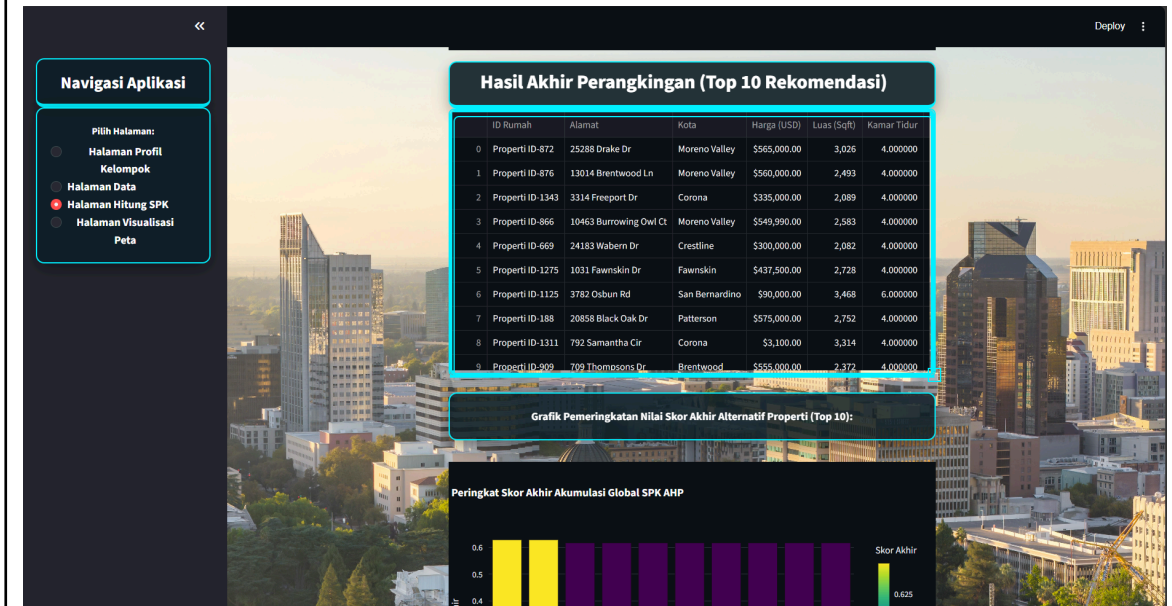
Picture 5 Pairwise, Normalize Matrix and Weight



Picture 6 Importance Visualization



Picture 7 Alternative Rank



Picture 8 Location Visualization



CHAPTER III

WORK SCHEDULE AND TASK DISTRIBUTION

3.1 Work Schedule

No	Activity	2026				
		Week				
		1	2	3	4	5
1	Literature Study	✓				
2	Dataset Collection	✓	✓			
3	System Development		✓	✓		
4	System Testing			✓	✓	
5	Report Writing			✓	✓	✓

3.2 Task Distribution

No	Name	SID	Deskripsi Tugas
1	Kevin Ridoi Parhusip	123240084	- Dataset collection - UI Streamlit - AHP implementation - Visualisasi peta Interaktif
2	Oktavian Prasetya Adi	123240135	- Dataset preprocessing - AHP process evaluation - Graph and Table visualization - Report Writing

CHAPTER IV

CONCLUSIONS AND SUGGESTIONS

4.1 Conclusion

Based on the final project results, the Analytical Hierarchy Process (AHP) method was successfully implemented in a web-based house selection Decision Support System. The system can assist users in determining the best house based on several criteria such as house price, building area, number of bedrooms, number of bathrooms, and house age.

The system can also automatically calculate priority weights and generate house rankings according to user preferences. With this system, the decision-making process becomes faster, more objective, and more efficient.

4.2 Suggestions

Suggestions for future system development include:

1. Adding more criteria such as location, garage, and environmental facilities.
2. Using a larger and real-time dataset.
3. Adding user login functionality.
4. Developing a mobile application version.
5. Combining AHP with other methods such as SAW or TOPSIS.

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